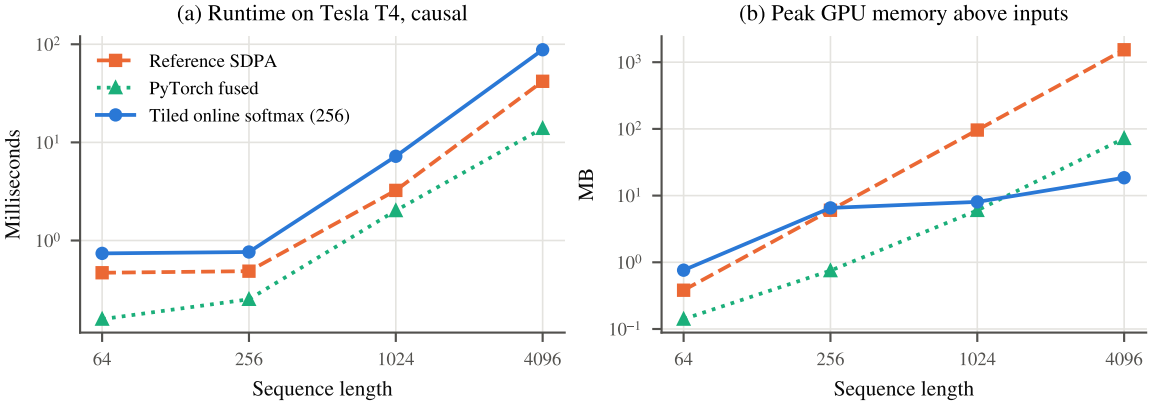
(a) Runtime on Tesla T4, causal
Reference SDPA
PyTorch fused
Tiled online softmax (256)
Milliseconds
10^2
10^1
10^0
64
256
1024
4096
Sequence length
(b) Peak GPU memory above inputs
MB
10^3
10^2
10^1
10^0
10^{-1}
64
256
1024
4096
Sequence length